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# MMA3001 | Numerical Methods and Machine Learning

## Week 06 Workshop

*Numerical Integration*


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# Attendance

| | | | | |
|---|---|---|---|---|
| **Workshop** | **Tuesday, 1 Sep** | **01** | **8:00AM** | **NWELN** |
| Tutorial | Thursday, 3 Sep | 01 | 12:00PM | S74N8 |

# Today's Workshop Agenda

- Summary of this Week's Learning
- Work together: Notebook Demonstrations

MMA3001 – Numerical Methods and Machine Learning

# Individual Project Brief

Key points for students to start considering the project

**25%**

of the unit mark

**Individual project**

Source: MMA3001_2026_Project_Brief.pdf

OVERVIEW

# What this project is assessing

Designed to assess the ability to apply computational techniques to meaningful engineering problems.

| | | |
|---|---|---|
| **Numerical methods** | **Machine learning** | **Computer vision** |
| **Data analysis** | **Optimisation** | **Combinations of techniques** |

## Assessment emphasis

- demonstrated individual learning
- quality and organisation of computational work
- validity of evaluation
- justification for technical decisions
- lessons drawn from results
- clear and professional communication

# Project selection and open-ended scope

## Selection route

- Select by the end of Week 6
- Choose a provided project and dataset
- Or propose an engineering problem and dataset
- Self-selected projects must be discussed before commencement

## Approval considers

- suitability for the unit
- clearly defined inputs and outputs
- achievable scope
- suitable validation evidence
- data ownership, licensing and privacy
- credible alternative solutions for comparison

*All projects are open-ended. The teaching team may ask for an overly complex proposal to be reduced or modified.*

SCOPE

# Possible project areas mentioned in the brief

## Apple-tree reconstruction

- Input: RGB-D observations of apple trees collected over three seasons
- Output: estimated three-dimensional tree skeletons
- Validation: comparison with ground-truth skeletons

## Frozen-meat packaging quality

- Input: RGB images of frozen-meat products and packaging
- Output: report describing meat quality and packaging quality
- Validation: comparison with labelled assessments or known inspection outcomes

## Simulation-based projects

- Students may work on simulations rather than data analysis
- Discuss this when proposing your own project
- Details of sample topics will be provided separately

CORE REQUIREMENTS

# Every project must include

## Engineering problem

Clearly defined and relevant to mechanical or aerospace engineering

## Technique

At least one numerical method and/or machine-learning technique

## Evidence

Model development; validation and verification; sensitivity analysis or optimisation where appropriate; engineering interpretation

## Workflow

Reproducible workflow using version control; GitHub required

## Tools

Appropriate use of Python, Jupyter, numerical libraries and ML frameworks

## AI use

Acknowledgement and details of AI usage

# Documentation: sections to build around

1 **Engineering problem**

context, problem, why it matters, intended use, limitations of scope

2 **Inputs and outputs**

input format, units and domain; outputs, meaning and intended use; invalid/missing inputs

3 **Computational solution**

method, reason selected, assumptions, settings or parameters, implementation, limitations

4 **Alternative solutions**

at least one alternative or baseline solution; evidence rather than preference alone

5 **Validation**

what was validated, appropriate method, metrics or acceptance criteria, and limitations

6 **Optimisation and performance**

accuracy, runtime, memory, robustness, stability, settings, resolution and trade-offs

7 **AI use and reflection**

AI tools, contribution, checking, student decisions, errors, limitations, understanding and workflow

# Computational approach and alternatives

## Explain the selected method

- how the method works at an appropriate level
- why it was selected
- important assumptions
- settings or parameters
- how it was implemented
- known limitations

## Compare alternatives or baselines

- accuracy, robustness and computational time
- memory use and implementation complexity
- interpretability and scalability
- suitability for the intended engineering application
- comparison is to show evidence-based selection

# Validation is required

Every project must include a clear and justified validation method.

- known ground truth
- independent experimental data
- trusted benchmark
- cross-validation
- sensitivity and robustness testing
- withheld test data
- repeated measurements
- established computational method
- physical constraints or conservation relationships
- manual inspection with defined criteria

**Explain: what was validated; why the method is appropriate; metrics or acceptance criteria; whether validation influenced model selection; and what validation cannot establish.**

*A numerical result or visually convincing output is not sufficient evidence of accuracy.*

# Optimisation and performance

## Analyse and improve where appropriate

- accuracy
- runtime
- memory requirements
- robustness
- numerical stability
- model or algorithm settings
- input resolution
- data requirements
- repeatability
- scalability
- trade-offs

## Use evidence to justify the final design

- make use of profilers and timing routines
- where appropriate, estimate arithmetic intensity and required FLOPS
- compare the original or baseline solution with the improved or selected solution
- justify the final design

# AI use and reflection

Students may use AI tools throughout the project, but all material AI use must be disclosed.

- which AI tools were used
- what they were used for
- approximate level of AI contribution
- why AI was used for those tasks
- how AI-generated material was checked
- which important decisions remained the student's responsibility
- errors, limitations or unhelpful suggestions produced by AI
- how AI use affected understanding or workflow

**Responsibility remains with the student**

Submitted code, results, claims, references and engineering decisions must still be verified.

*AI cannot be used to answer presentation and interview questions.*

DELIVERABLE

# GitHub repository: required contents

### README file

orientation and workflow instructions

### HTML source-code documentation

can be generated by pdoc or similar

### Licensing agreement

included in the repository

### Source code

functional main branch with inline comments and docstrings

### Test functions and test report

can be generated by pytest

### Accessible final link and zipped clone

link submitted; zipped clone submitted via Moodle

*The repository must contain the code and documentation needed to review the computational work.*

TIMELINE

# Assessment deliverables and timing

1

**End of Week 6**

Select project after the mid-semester assessment

2

**Remaining teaching weeks**

Expected progress and project guidance discussed

3

**Due dates**

See Unit Overview Spreadsheet

4

**Week 14 examination period**

Individual presentation / demonstration

Written documentation: individual document of 5–10 pages maximum; references and appendices do not count towards the 10-page limit; submit via Moodle.

GitHub: accessible final repository link plus a zipped clone submitted via Moodle.

# Presentation and interview

**5 min**

presentation

**3 min**

questions

Week 14 during the examination period

## Presentation should include

- brief engineering problem: primary input and output
- GitHub and structure; code documentation
- code validation and testing evidence
- chosen computational approach and reason for choosing it
- principal limitations or lesson learned
- student use and evaluation of AI

## Question themes may include

- represented physical or engineering system and assumptions
- influential parameters or boundary conditions
- numerical stability and convergence criteria
- sources of numerical error
- verification against analytical, manufactured or benchmark solutions
- vulnerability to numerical failure and mitigation

ASSESSMENT FOCUS

# What strong work demonstrates

### Problem & I/O

complete engineering context; inputs and outputs fully specified with format, units, domains, meaning and intended use

### Repository

professional, clear, fully reproducible structure; README with complete orientation and workflow instructions

### Code documentation

readable code; comments clarify intent; consistent docstrings; complete HTML documentation

### Validation

appropriate validation design, correct metrics, comparisons and honest limitations

### Approach justification

credible baseline, evidence-based method selection, quantitative analysis and trade-offs

### AI reflection

complete disclosure, verification of AI content, identification of errors and clear evidence of understanding

*A perfect solution is not required: a partially successful method can still demonstrate strong engineering work if it is validated honestly and its limitations, lessons and improvements are clearly explained.*

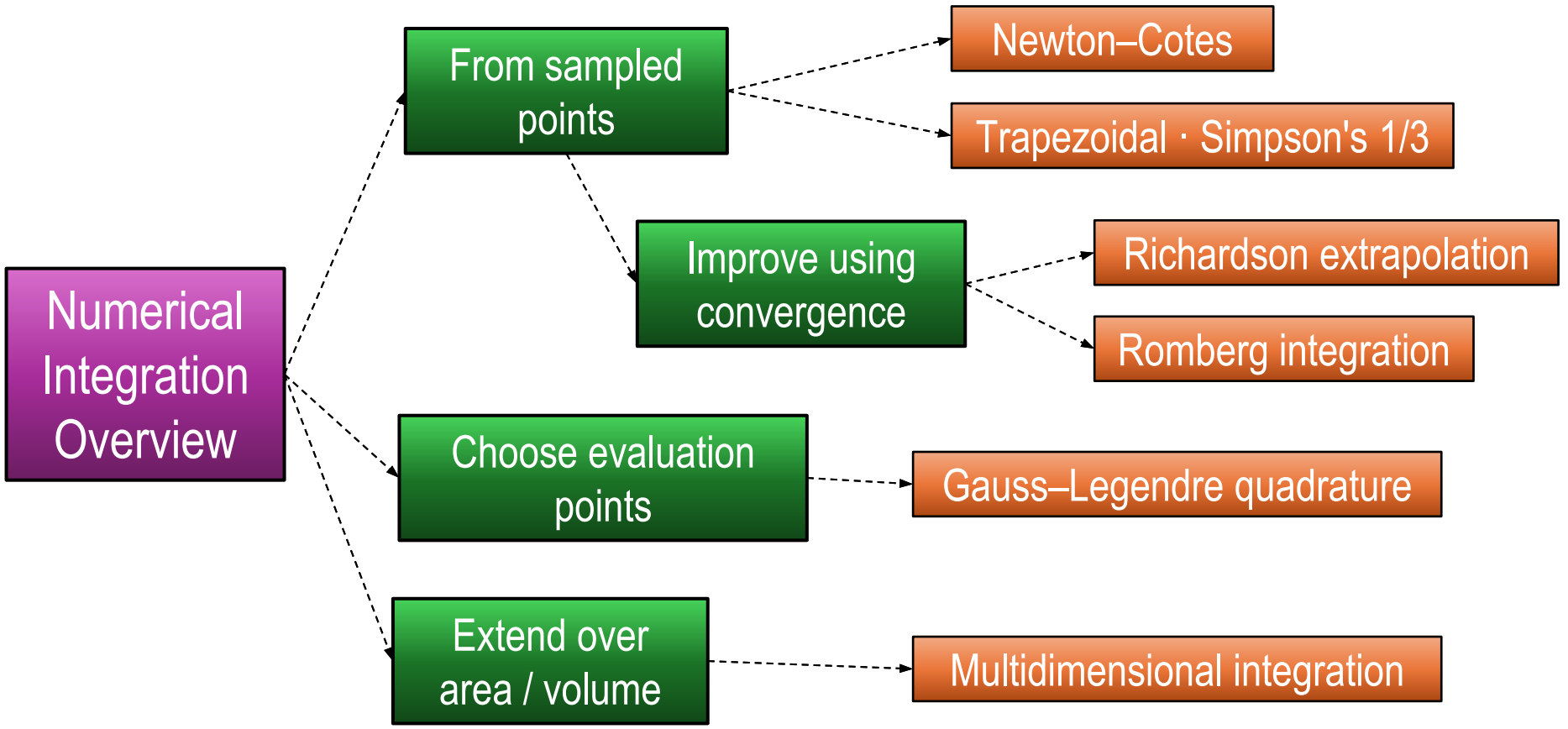
# Possible Open-Source Engineering Datasets

Dataset names and link labels are clickable. Students must still define a suitable engineering problem, inputs/outputs, method, baseline and validation plan.

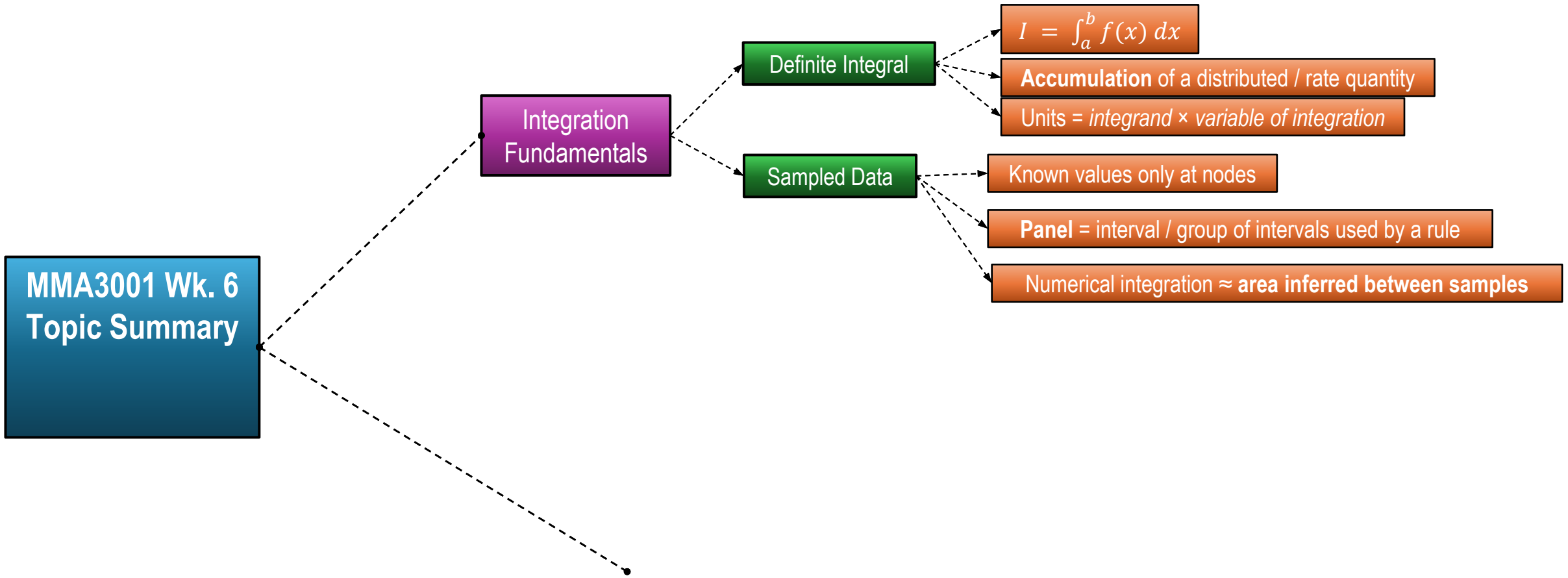
| Dataset | Possible engineering use | Source | Working link |
|---|---|---|---|
| Airfoil Self-Noise | Aeroacoustics / regression | UCI 291 | Open official page |
| Combined Cycle Power Plant | Thermodynamics / power output | UCI 294 | Open official page |
| Gas Turbine CO and NOx Emissions | Gas turbine emissions / energy yield | UCI 551 | Open official page |
| Hydraulic System Condition Monitoring | Fluid power / component condition | UCI 447 | Open official page |
| AI4I Predictive Maintenance | Manufacturing / machine failure | UCI 601 | Open official page |
| Energy Efficiency | Building thermal load prediction | UCI 242 | Open official page |
| Steel Plates Faults | Manufacturing quality classification | UCI 198 | Open official page |
| MetroPT-3 | Rail compressor predictive maintenance | UCI 791 | Open official page |
| NASA C-MAPSS Jet Engine Simulated Data | Turbofan RUL / prognostics | NASA | Open official page |

*Note: NASA C-MAPSS is publicly accessible via NASA Open Data; UCI datasets above are listed on the UCI Machine Learning Repository.*

# Week 6 Summary (Overview)

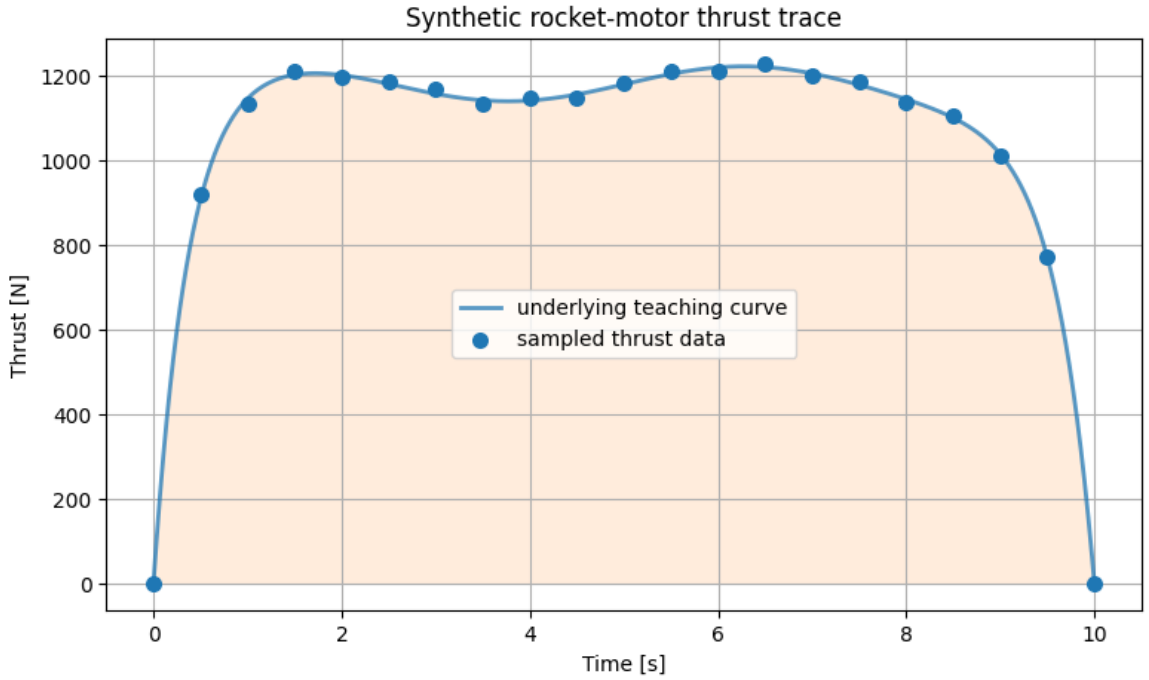


# Week 6 Summary (Part 1)



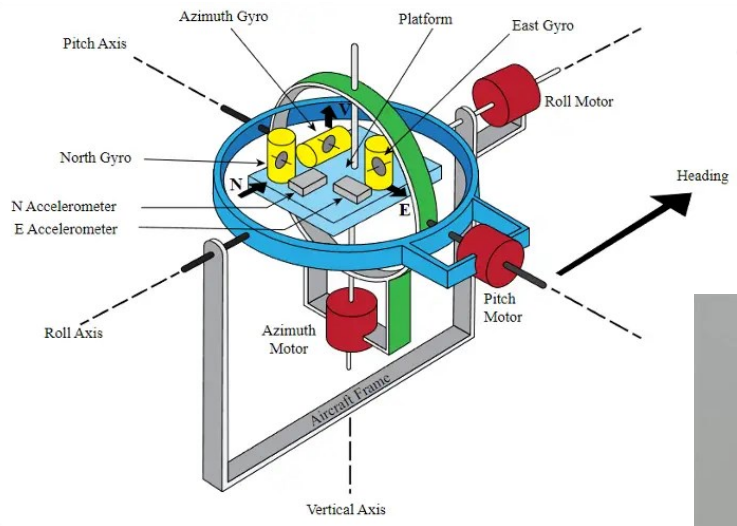
# Week 6 Summary (Part 1)

- **Integration Fundamentals**
  - **Definite Integral**
    - $I = \int_a^b f(x)\,dx$
    - **Accumulation** of a distributed / rate quantity
    - Units = *integrand* × *variable of integration*
  - **Sampled Data**
    - Known values only at nodes
    - **Panel** = interval / group of intervals used by a rule
    - Numerical integration ≈ **area inferred between samples**




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# Week 6 Summary (Part 1)




https://www.dextermag.com/wp-content/uploads/2022/01/accelerometer1.webp




https://aviationweek.com/air-transport/aircraft-propulsion/airbus-a319neo-concludes-maiden-flight



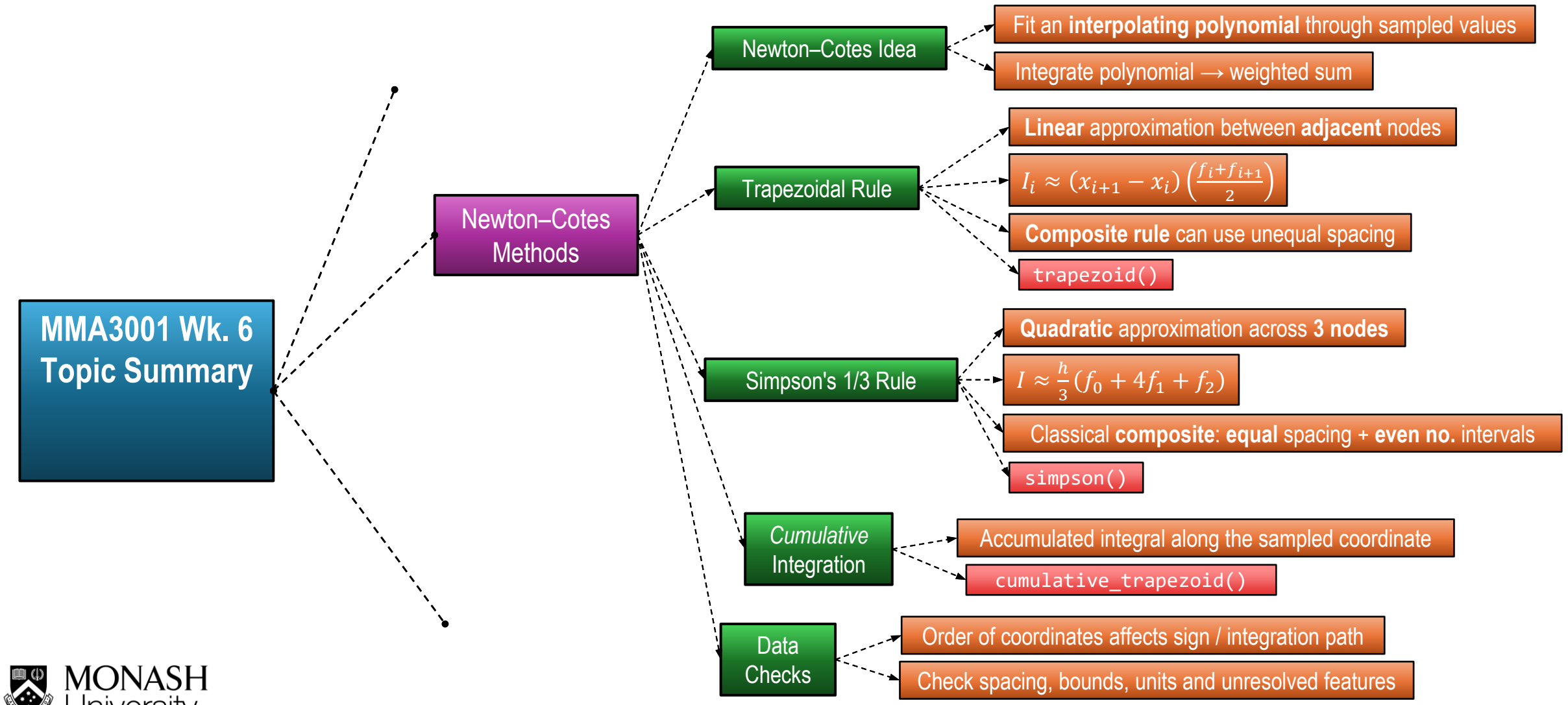
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https://my.element14.com/schneider-electric/metsepm5110/power-meter-3-ph-100-415vac-panel/dp/2835421

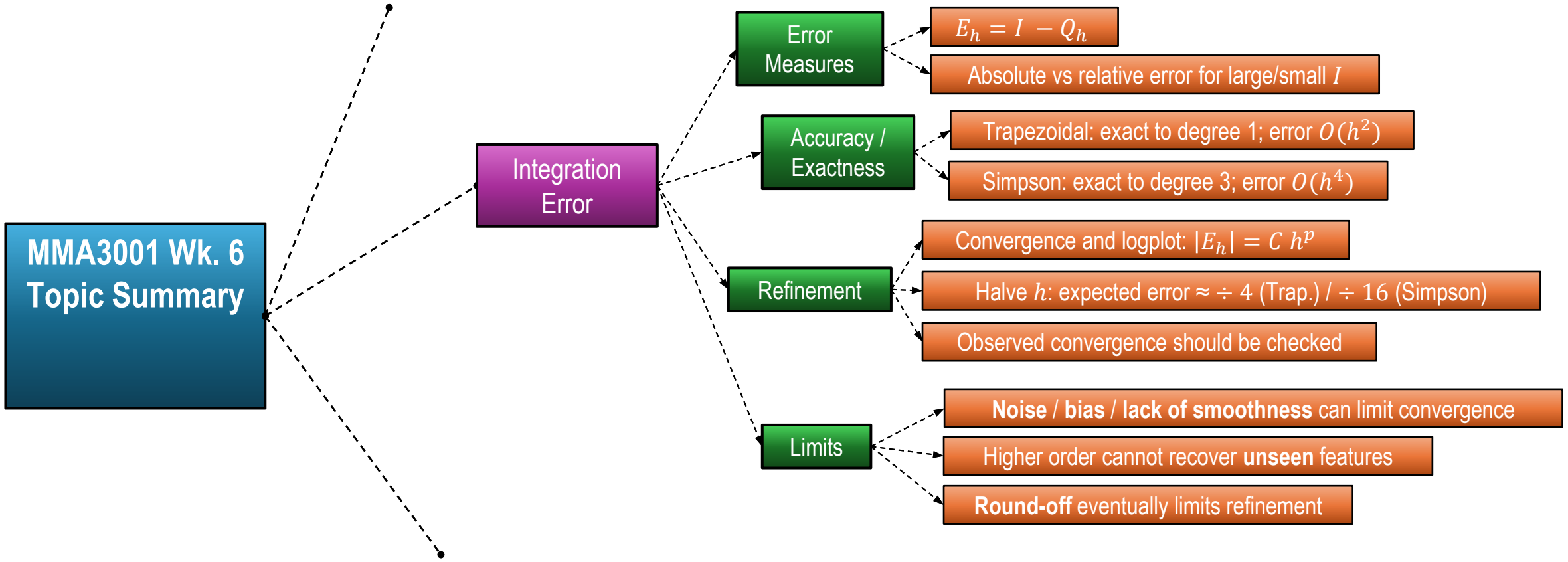
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# Week 6 Summary (Part 2)



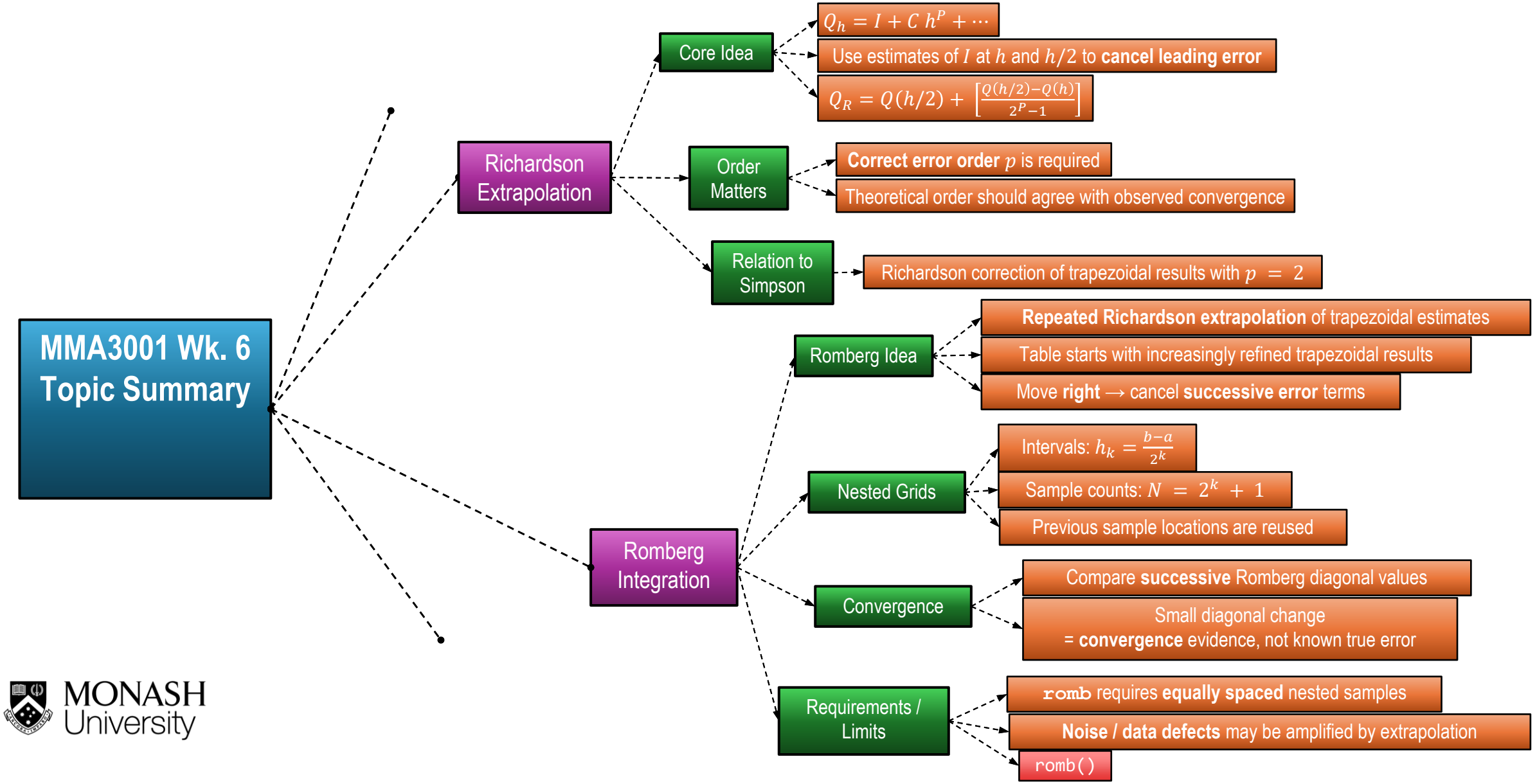

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# Week 6 Summary (Part 3)



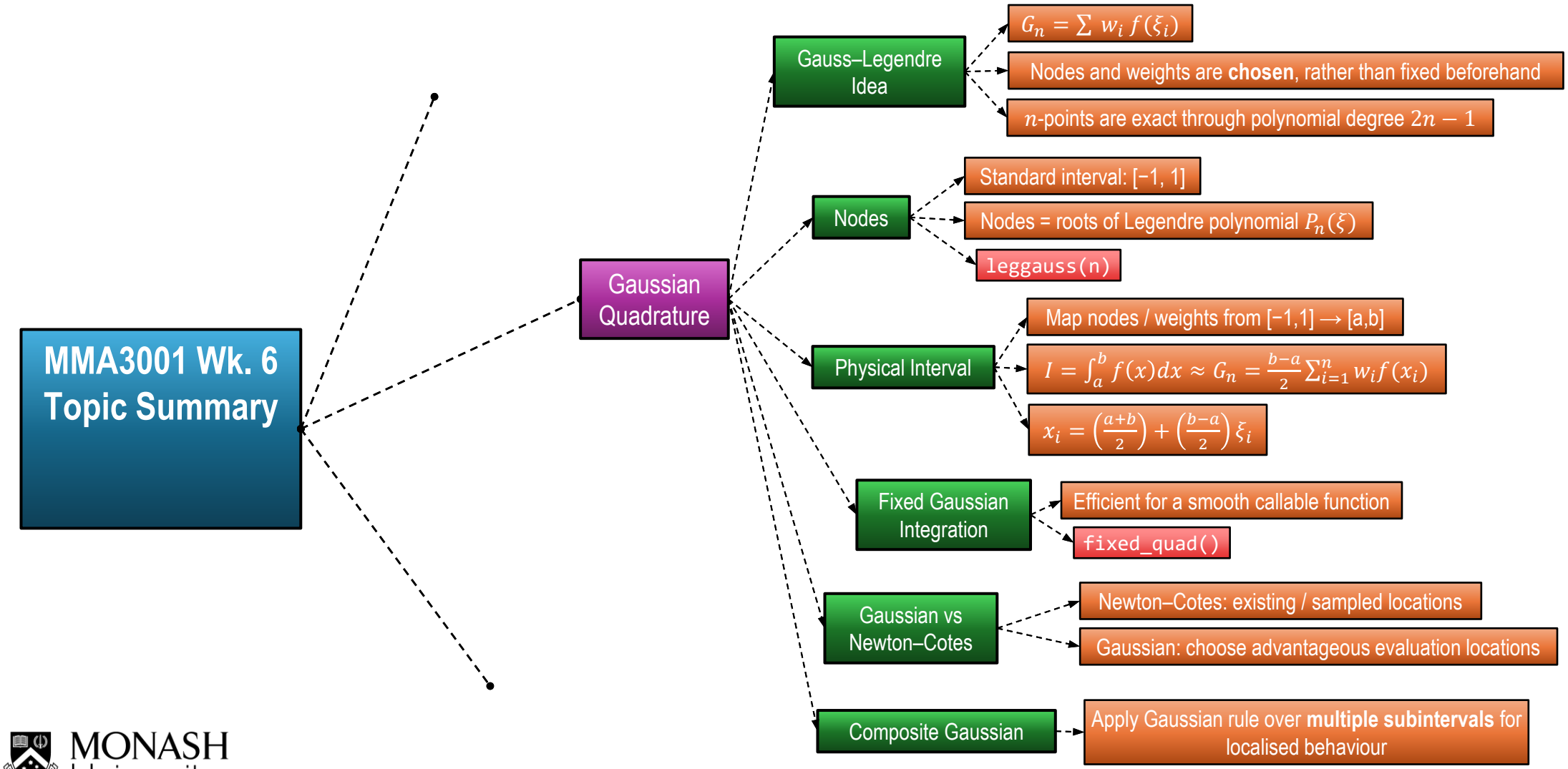

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# Week 6 Summary (Part43)



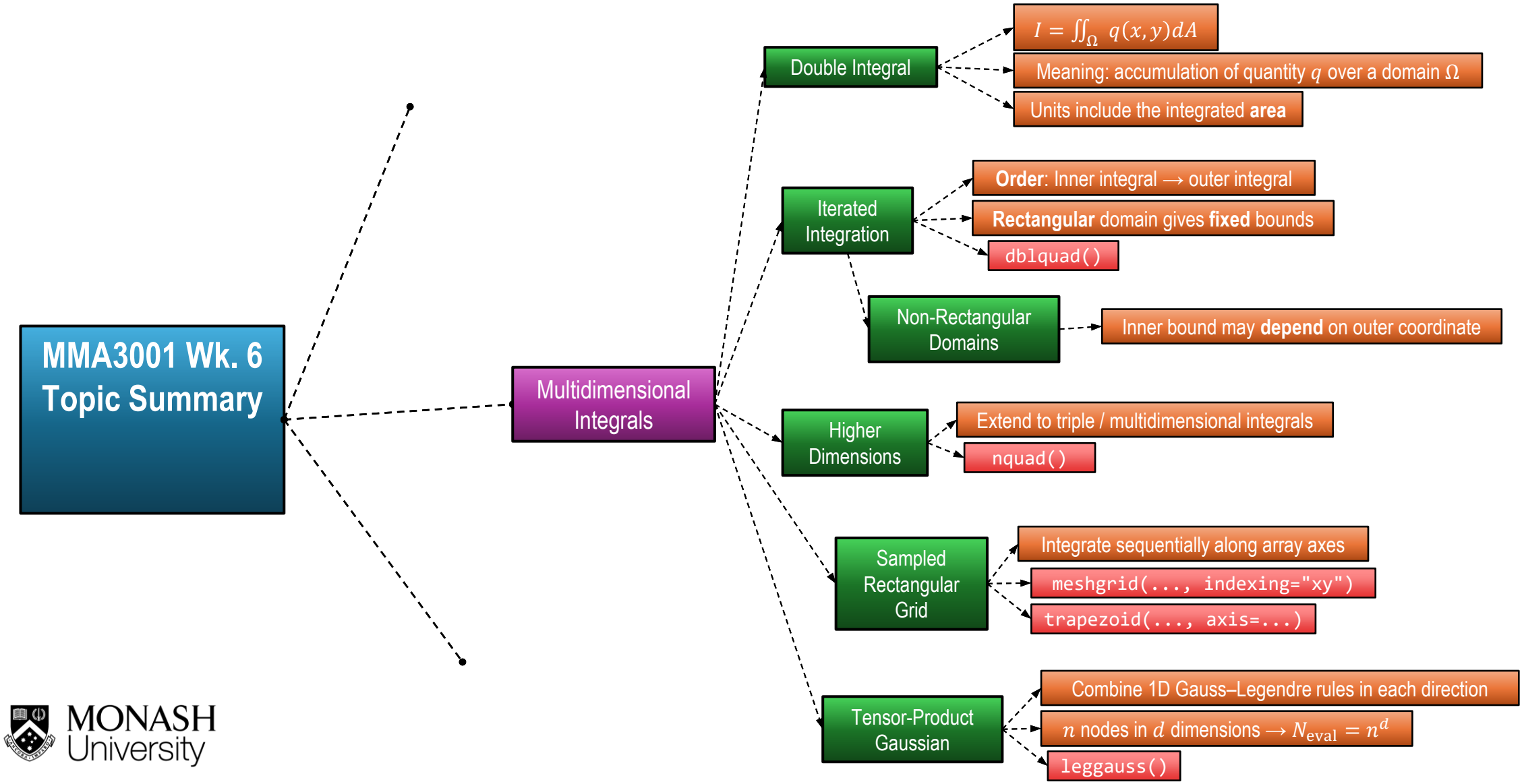

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# Week 6 Summary (Part 5)



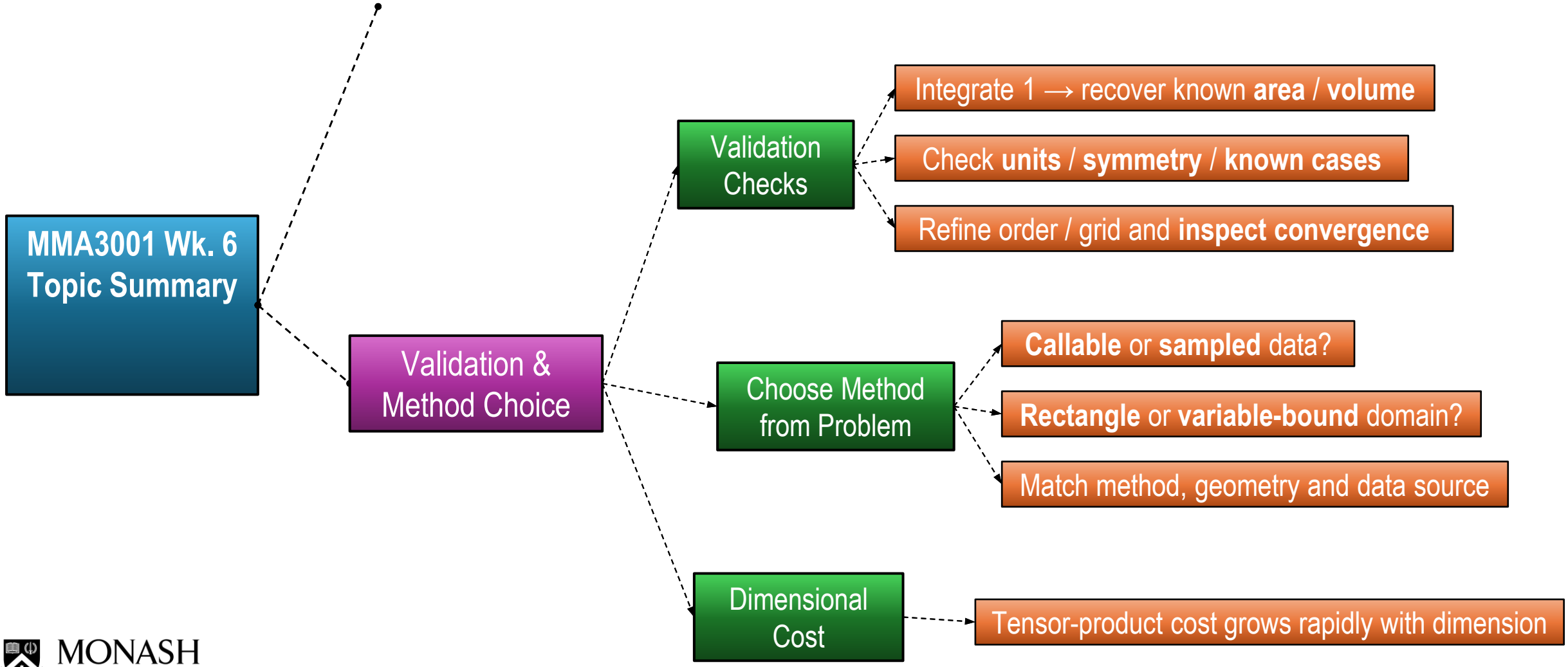

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# Week 6 Summary (Part 6)




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# Week 6 Summary (Part 7)




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# Practical Advice for Thursday

## Before class

- Review the six Week 5 notebooks and workshop material.
- You are not expected to complete the workbook beforehand.
- Use the Overview Sheet if you need to revisit a topic or recording.

## During class

- Work through Activities 1–6 using: Predict → Run → Explain → Test
  - **Predict** what the code will do.
  - **Run** and inspect the result.
  - **Explain** the engineering meaning.
  - Change a value and **test** your understanding.

# AI Acknowledgement

LLM models (OpenAI's ChatGPT 5.6, Google Gemini 2.5) were used in the creation of these slides to check grammar, clarity of phrasing, and generate icons and markdown elements for the layout.

All content has been reviewed, verified, and adapted by the author to ensure accuracy, clarity, and alignment with the learning objectives.